Web Search and Mining (WSM) - Information Retrieval Reference Notes

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Information Retrieval (IR) is the discipline concerned with indexing, searching, and discovering knowledge within unstructured collections of documents.

Key Concepts in Search Engine Architecture:

1. Positional Inverted Index:
The positional inverted index maps every distinct vocabulary term to a postings list containing document identifiers and the exact sequential token offsets:
    Term -> { DocumentID: [pos1, pos2, pos3, ...] }
This structure enables efficient exact phrase queries ("fried chicken", "civil war", "house divided") and proximity searches without re-parsing raw files.

2. Boolean Retrieval Model:
Executes classic propositional logic queries over document sets:
- AND: Computes set intersection of postings lists.
- OR: Computes set union of postings lists.
- NOT: Computes set difference to exclude unwanted topics.

3. Wildcard Query Processing:
Wildcard patterns containing * or ? (such as napoleon*, *ing, m*n) are converted to compiled regular expressions and evaluated against the indexed vocabulary, merging document postings across matching terms.

4. TF-IDF Relevance Scoring & Ranking:
To rank matching documents by relevance, the system calculates Term Frequency - Inverse Document Frequency weights:
    Score = (1 + log10(TF)) * log10((N + 1) / (DF + 1))
Scores are adjusted with cosine length normalization to ensure fair competition between short and long documents.

5. Query Expansion:
Leverages lexical databases like NLTK WordNet and curated domain lexicons to propose relevant synonyms and related concepts, improving recall for user discovery.